



SkyStitch

Manual & Workflow Guide

Bilingual • English / Bahasa Indonesia

v2.3 • 2D orthomosaic + 3D on a local node (NodeODM) • 3 August 2026

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How to read: *One bilingual file: each part is written first in English, then in Bahasa Indonesia. Jump via the headings.*

Part 1 — SkyStitch Plugin Manual

ENGLISH

SkyStitch is a QGIS plugin that turns raw drone photos into maps. The menu (Raster → SkyStitch) shows two tools:

- **Drone Orthomosaic (2D):** a fast 2D stitch (GPS + SIFT homography) that outputs one georeferenced orthomosaic (GeoTIFF). Best for flat terrain and quick previews. No engine needed.
- **3D Photogrammetry (local node):** full photogrammetry on a NodeODM processing node running on your own machine, returning a true orthophoto plus DSM, DTM/DEM, point cloud and 3D mesh — loaded straight into QGIS.

Advanced: Two more engines (WebODM with login, and local OpenDroneMap via Docker) still ship but are hidden from the menu; enable them by flipping `SHOW_WEBODM_REMOTE_ENGINE` / `SHOW_LOCAL_DOCKER_ENGINE` in `skystitch_plugin.py`.

1. Install the plugin

- **From ZIP:** In QGIS: Plugins → Manage and Install Plugins → Install from ZIP → choose the SkyStitch .zip → Install Plugin, then restart QGIS.
- **Manual:** Copy the skystitch folder into your QGIS profile's python/plugins/ folder (Settings → User Profiles → Open Active Profile Folder), then enable it in Plugins → Installed.

After install, two entries appear in the Raster → SkyStitch menu and two toolbar buttons.

2. Requirements

2D engine — Python packages

The 2D engine needs a few packages in QGIS's own Python. Open the OSGeo4W Shell (Windows) and run, then restart QGIS:

```
python-qgis -m pip install opencv-python-headless exifread rasterio pyproj scipy pillow
```

3D engine — a local NodeODM node (install once)

The 3D engine does the heavy photogrammetry on a NodeODM node. High-accuracy 3D always needs such an engine — it cannot live inside the plugin — so each user installs it once. After that, everything runs from QGIS: no Docker to launch, no cloud.

1. Install a NodeODM engine. Easiest on Windows: the free WebODM native installer (webodm.org/download) — it runs natively (no Docker) and provides a local node. A standalone NodeODM works too.
2. Install the requests package into QGIS's Python (once):

```
python-qgis -m pip install requests
```
3. Find the node's address: open the WebODM app → Processing Nodes → note the Hostname/Port (e.g. `http://localhost:29301`). The native installer's port can change between launches, so check it here each time.

3. Using the 2D engine

1. Open Raster → SkyStitch → SkyStitch - Drone Orthomosaic (2D).
2. Select the drone photo folder (JPG with GPS EXIF) and set the output .tif file.
3. Optional: limit photo count for a quick test, set GSD, choose compression, or add a GCP/ICP file (CSV/XLSX).
4. Click Build Mosaic. It runs in the background; progress and log show in the dialog and it can be canceled.
5. When finished, the orthomosaic (single GeoTIFF) loads into the canvas.

Tip: Overlap $\geq 60\text{--}70\%$ is recommended. Photos that don't overlap are still placed at their own GPS position, not dropped.

4. Using the 3D engine (local node)

Make sure the WebODM app (or your NodeODM) is running — that is the node; if it is closed, the connection is refused.

1. Open Raster → SkyStitch → SkyStitch - 3D Photogrammetry (local node).
2. Enter the Node URL from Processing Nodes (e.g. <http://localhost:29301>); add a token only if your node requires one. Click Test Connection — it should say OK.
3. Pick the drone photo folder and an output folder.
4. Choose products (DSM / DTM / mesh), point-cloud quality (start with Medium), orthophoto resolution, and an optional GCP list.
5. Click Process on node. Photos upload (resumable), process on the local node, then the results are downloaded, extracted, and loaded into the canvas. Progress/log stream; the job can be canceled.

Outputs (extracted into <output>/results/)

- **Orthophoto** — odm_orthophoto/odm_orthophoto.tif
- **DSM / DTM** — odm_dem/dsm.tif · odm_dem/dtm.tif
- **Point cloud** — odm_georeferencing/odm_georeferenced_model.laz
- **3D mesh** — odm_texturing/odm_textured_model_geo.obj

5. Which engine should I use?

Aspect	2D engine	3D engine (local node)
Method	SIFT + homography stitch	SfM/MVS photogrammetry (NodeODM)
Outputs	1 orthomosaic (GeoTIFF)	ortho + DSM + DTM + point cloud + mesh
Elevation (DSM/DEM)	No	Yes
Terrain	Flat only	Any (hills, buildings)
Setup	Python packages	Install a NodeODM engine once + requests
Runs on	This machine, in QGIS	Local node, driven from QGIS
Best for	Quick 2D preview	Survey-grade, elevation, 3D

6. Troubleshooting

- **3D — “Connection failed / refused”:** (1) make sure the WebODM app / NodeODM is running; (2) use the exact Node URL & port from WebODM → Processing Nodes — the native installer's port can change each launch; (3) if the dialog mentions requests, run `python-qgis -m pip install requests` and restart QGIS.
- **3D — “Not enough photos”:** the node needs several overlapping photos (typically 20+). For 1–2 photos, use the 2D engine.
- **2D — missing modules / PROJ error:** run the pip install into QGIS's own Python (section 2), then fully restart QGIS. The bundled `proj_fix.py` corrects the common PROJ `proj.db` version error.

Bagian 1 — Manual Plugin SkyStitch

BAHASA INDONESIA

SkyStitch adalah plugin QGIS untuk mengubah foto drone mentah menjadi peta. Di menu (Raster → SkyStitch) ada dua tool:

- **Drone Orthomosaic (2D):** stitch 2D cepat (GPS + homografi SIFT) yang menghasilkan satu orthomosaic bergeoreferensi (GeoTIFF). Cocok untuk medan datar & pratinjau cepat. Tanpa mesin tambahan.
- **3D Photogrammetry (local node):** fotogrametri penuh di node NodeODM yang berjalan di PC-mu sendiri, menghasilkan orthophoto sejati plus DSM, DTM/DEM, point cloud, dan mesh 3D — langsung dimuat ke QGIS.

Lanjutan: Dua mesin lain (WebODM dengan login, dan OpenDroneMap lokal via Docker) tetap disertakan tapi disembunyikan dari menu; aktifkan lewat `SHOW_WEBODM_REMOTE_ENGINE` / `SHOW_LOCAL_DOCKER_ENGINE` di `skystitch_plugin.py`.

1. Memasang plugin

- **Dari ZIP:** Di QGIS: Plugins → Manage and Install Plugins → Install from ZIP → pilih file .zip SkyStitch → Install Plugin, lalu restart QGIS.
- **Manual:** Salin folder skystitch ke folder python/plugins/ pada profil QGIS (Settings → User Profiles → Open Active Profile Folder), lalu aktifkan di Plugins → Installed.

Setelah terpasang, ada dua entri di menu Raster → SkyStitch dan dua tombol toolbar.

2. Prasyarat

Mesin 2D — paket Python

Mesin 2D butuh beberapa paket di Python milik QGIS. Buka OSGeo4W Shell (Windows), jalankan, lalu restart QGIS:

```
python-qgis -m pip install opencv-python-headless exifread rasterio pyproj scipy pillow
```

Mesin 3D — node NodeODM lokal (pasang sekali)

Mesin 3D memproses fotogrametri berat di node NodeODM. Fotogrametri 3D akurat SELALU butuh mesin seperti ini — tidak bisa hidup di dalam plugin — jadi tiap pengguna memasangnya sekali. Setelahnya, semua dijalankan dari QGIS: tanpa membuka Docker, tanpa cloud.

1. Pasang mesin NodeODM. Termudah di Windows: WebODM native installer gratis (webodm.org/download) — jalan native (tanpa Docker) dan menyediakan node lokal. NodeODM standalone juga bisa.
2. Pasang paket requests ke Python QGIS (sekali):

```
python-qgis -m pip install requests
```
3. Temukan alamat node: buka aplikasi WebODM → Processing Nodes → catat Hostname/Port (mis. `http://localhost:29301`). Port installer native bisa berubah tiap kali dibuka, jadi cek di sini setiap saat.

3. Memakai mesin 2D

1. Buka Raster → SkyStitch → SkyStitch - Drone Orthomosaic (2D).
2. Pilih folder foto drone (JPG dengan GPS EXIF) dan tentukan berkas keluaran .tif.
3. Opsional: batasi jumlah foto untuk uji cepat, atur GSD, pilih kompresi, atau tambahkan berkas GCP/ICP (CSV/XLSX).
4. Klik Build Mosaic. Berjalan di latar belakang; progres & log tampil di dialog dan bisa dibatalkan.
5. Setelah selesai, orthomosaic (satu GeoTIFF) dimuat ke kanvas.

Tips: Overlap $\geq 60-70\%$ disarankan. Foto yang tidak beroverlap tetap ditempatkan pada posisi GPS-nya sendiri, bukan dibuang.

4. Memakai mesin 3D (local node)

Pastikan aplikasi WebODM (atau NodeODM-mu) sedang berjalan — itu node-nya; kalau ditutup, koneksi akan ditolak (refused).

1. Buka Raster → SkyStitch → SkyStitch - 3D Photogrammetry (local node).
2. Isi Node URL dari Processing Nodes (mis. <http://localhost:29301>); tambahkan token hanya bila node-mu memerlukannya. Klik Test Connection — harus muncul OK.
3. Pilih folder foto drone dan folder output.
4. Pilih produk (DSM / DTM / mesh), kualitas point cloud (mulai Medium), resolusi orthophoto, dan GCP opsional.
5. Klik Process on node. Foto ter-upload (resumable), diproses di node lokal, lalu hasil diunduh, diekstrak, dan dimuat ke kanvas. Progres/log tampil; bisa dibatalkan.

Output (diekstrak ke <output>/results/)

- **Orthophoto** — odm_orthophoto/odm_orthophoto.tif
- **DSM / DTM** — odm_dem/dsm.tif · odm_dem/dtm.tif
- **Point cloud** — odm_georeferencing/odm_georeferenced_model.laz
- **Mesh 3D** — odm_texturing/odm_textured_model_geo.obj

5. Pilih mesin yang mana?

Aspek	Mesin 2D	Mesin 3D (local node)
Metode	Stitch SIFT + homografi	Fotogrametri SfM/MVS (NodeODM)
Output	1 orthomosaic (GeoTIFF)	ortho + DSM + DTM + point cloud + mesh
Ketinggian (DSM/DEM)	Tidak	Ya
Medan	Hanya datar	Apa saja (bukit, bangunan)
Setup	Paket Python	Pasang mesin NodeODM sekali + requests
Jalan di	PC ini, di QGIS	Node lokal, dikendalikan dari QGIS
Cocok untuk	Pratinjau 2D cepat	Survey-grade, ketinggian, 3D

6. Pemecahan masalah

- **3D — “Connection failed / refused”:** (1) pastikan aplikasi WebODM / NodeODM sedang jalan; (2) pakai Node URL & port PERSIS dari WebODM → Processing Nodes — port installer native bisa berubah tiap dibuka; (3) bila dialog menyebut requests, jalankan `python-qgis -m pip install requests` lalu restart QGIS.
- **3D — “Not enough photos”:** node butuh beberapa foto beroverlap (umumnya 20+). Untuk 1–2 foto, pakai mesin 2D.
- **2D — modul hilang / error PROJ:** jalankan `pip install` ke Python milik QGIS (bagian 2), lalu restart QGIS sepenuhnya. `proj_fix.py` bawaan memperbaiki error versi PROJ `proj.db` yang umum.

Part 2 — Drone Data Processing Workflow (WebODM + QGIS)

ENGLISH

A 9-step workflow equivalent to the ArcGIS Pro drone poster, but simpler (WebODM automates the heavy photogrammetry) and more accurate than 2D stitching (real SfM/MVS, GCP-ready). All tools are free and open-source. This mirrors what the plugin's 3D engine does under the hood.

Requirements

- **Engine** — WebODM (native installer or a NodeODM node). Free.
- **QGIS 3.x** — to display, check and derive analysis products. Free.
- **Photos** — JPG drone photos with GPS EXIF; overlap $\geq 70\%$ (front) & $\geq 60\%$ (side).
- **Computer** — RAM ≥ 8 GB (16 GB+ recommended), GPU helps, plenty of disk.
- **Optional** — surveyed Ground Control Points (GCP) for absolute accuracy.

The 9 steps

1. Data preparation

Collect all photos into one folder. Ensure enough overlap ($\geq 70\%$ front, $\geq 60\%$ side) and GPS EXIF. Decide the target CRS (e.g. WGS 84 / UTM Zone 48S for Java). Prepare a GCP file if available.

2. New project (in WebODM)

Open WebODM → Add Project → name it. No manual CRS setup — the engine picks the UTM zone from photo GPS.

3. Add photos

Inside the project → Select Images and GCP → pick all photos (drag-drop). Include gcp_list.txt here if using GCP.

4. Set up photogrammetry (options & GCP)

Enable DSM, DTM, point-cloud quality and orthophoto resolution. With GCP loaded, absolute accuracy improves to $\sim 2.5 \times \text{GSD}$ horizontal, $4 \times \text{GSD}$ vertical.

5. Process (Build & Process, automatic)

Click Start. The engine runs the whole chain automatically: feature matching & SfM → dense cloud (MVS) → meshing & texturing → DSM/DTM → orthophoto.

6. Main outputs

Orthophoto (odm_orthophoto.tif), DSM (dsm.tif), DTM/DEM (dtm.tif), point cloud (.laz), textured 3D mesh (.obj).

7. Display & check (in QGIS)

Load the orthophoto and DSM into QGIS. Check position, completeness and distortion; style the DSM. Read the quality report (GSD, reprojection error, GCP/checkpoint accuracy).

8. Analysis & derivatives (in QGIS)

Contour: Raster ► Extraction ► Contour. Slope/Aspect/Hillshade: Raster ► Analysis. Cut/Fill: Raster Calculator difference of two DSMs, then Raster Surface Volume. Land cover: SCP or dzetsaka plugin on the orthophoto.

9. Export & share

Export GeoTIFF (ortho/DSM), LAS/LAZ (point cloud), Shapefile/GeoPackage (contours, classification). Share as a layer, map package, or web map (qgis2web/QField/GeoServer).

Tips

- **Overlap:** photos $\geq 70\%$ front & $\geq 60\%$ side.
- **GCP:** use GCP for absolute accuracy; 8–12 photos per GCP is enough, keep some as checkpoints.
- **Performance:** RAM & GPU strongly affect time; large sets take hours. Test with a subset first.
- **Backup:** back up data regularly.

Bagian 2 — Alur Pengolahan Data Drone (WebODM + QGIS)

BAHASA INDONESIA

Alur 9 langkah setara poster drone ArcGIS Pro, tetapi lebih simpel (WebODM mengotomatiskan fotogrametri berat) dan lebih akurat daripada penjahitan 2D (SfM/MVS sungguhan, siap GCP). Semua alat gratis & open-source. Ini mencerminkan yang dikerjakan mesin 3D plugin di balik layar.

Prasyarat

- **Mesin** — WebODM (installer native atau sebuah node NodeODM). Gratis.
- **QGIS 3.x** — untuk menampilkan, memeriksa, dan menurunkan produk analisis. Gratis.
- **Foto** — foto drone JPG dengan GPS EXIF; overlap $\geq 70\%$ (depan) & $\geq 60\%$ (samping).
- **Komputer** — RAM ≥ 8 GB (16 GB+ disarankan), GPU membantu, ruang disk lega.
- **Opsional** — titik kontrol tanah (GCP) hasil ukur untuk akurasi absolut.

9 langkah

1. Persiapan data

Kumpulkan semua foto ke satu folder. Pastikan overlap cukup ($\geq 70\%$ depan, $\geq 60\%$ samping) dan ada GPS EXIF. Tentukan CRS target (mis. WGS 84 / UTM Zona 48S untuk Jawa). Siapkan berkas GCP bila ada.

2. Proyek baru (di WebODM)

Buka WebODM → Add Project → beri nama. Tak perlu atur CRS manual — mesin memilih zona UTM dari GPS foto.

3. Tambahkan foto

Di dalam proyek → Select Images and GCP → pilih semua foto (drag-drop). Sertakan gcp_list.txt bila memakai GCP.

4. Siapkan fotogrametri (opsi & GCP)

Aktifkan DSM, DTM, kualitas point cloud, dan resolusi orthophoto. Dengan GCP, akurasi absolut membaik hingga $\pm 2,5 \times \text{GSD}$ horizontal, $4 \times \text{GSD}$ vertikal.

5. Proses (Build & Process, otomatis)

Klik Start. Mesin menjalankan seluruh rantai otomatis: pencocokan fitur & SfM → dense cloud (MVS) → meshing & texturing → DSM/DTM → orthophoto.

6. Hasil utama

Orthophoto (odm_orthophoto.tif), DSM (dsm.tif), DTM/DEM (dtm.tif), point cloud (.laz), mesh 3D bertekstur (.obj).

7. Tampilkan & periksa (di QGIS)

Muat orthophoto dan DSM ke QGIS. Periksa posisi, kelengkapan, dan distorsi; atur symbology DSM. Baca laporan kualitas (GSD, reprojection error, akurasi GCP/checkpoint).

8. Analisis & produk turunan (di QGIS)

Kontur: Raster ► Extraction ► Contour. Slope/Aspect/Hillshade: Raster ► Analysis. Cut/Fill: selisih dua DSM di Raster Calculator, lalu Raster Surface Volume. Tutupan lahan: plugin SCP atau dzetsaka pada orthophoto.

9. Ekspor & berbagi

Ekspor GeoTIFF (ortho/DSM), LAS/LAZ (point cloud), Shapefile/GeoPackage (kontur, klasifikasi). Bagikan sebagai layer, paket peta, atau peta web (qgis2web/QField/GeoServer).

Tips

- **Overlap:** foto $\geq 70\%$ depan & $\geq 60\%$ samping.
- **GCP:** pakai GCP untuk akurasi absolut; 8–12 foto per GCP cukup, sisakan beberapa sebagai checkpoint.
- **Performa:** RAM & GPU sangat memengaruhi waktu; dataset besar berjam-jam. Uji sebagian dulu.
- **Cadangan:** backup data secara berkala.

References / Referensi

- [OpenDroneMap — Outputs](#)
- [OpenDroneMap — Map accuracy & GCP](#)
- [WebODM — Download \(native, no Docker\)](#)
- [QGIS — Terrain Analysis](#)